Joshua Zatz

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EDUCATION

University of Illinois Chicago Chicago, IL
Master of Science Expected May 2027
Major in Mechanical Engineering; Concentration in Robotics
Cumulative GPA: 4.0/4.0; Dean's List 2015-2016

University of Illinois Chicago Chicago, IL
Bachelor of Science Aug 2020 - May 2025
Major in Mechanical Engineering
Cumulative GPA: 3.8/4.0; Magna Cum Laude
Extracurricular: Co-Founder of ASME @ UIC, Active Member of SAE

WORK EXPERIENCE

Abbott Laboratories | Contract Job Chicago, IL
Mechanical Engineer Sept 2024 – May 2025
- Developed an RFID sensor to monitor medical supplies dynamically when placed outside a temperature range, which reduced the cost of current sensors and risk due to spoiled supplies.
- Performed CFD simulations to determine key points in the model that can be tuned for different temperature-dependent medical supplies.
- Prototyped and tested different medical-grade porous films that result in the least amount of error with minimised size to reduce cost.

United States Gypsum | Internship Chicago, IL
Thermal Engineer May 2023 – Aug 2023
- Collaborated in the analysis, design and implementation of a heat exchanger system that led to 28MBTU/year saved and significant cost reductions after the 6-year ROI.
- Created a system to measure and test moisture in the factories' mills, effectively reducing the gypsum calcination time by ~5%
- Performed Factory Acceptance Tests (FATs) on newly manufactured burners and mills.

PROJECTS

FermiLab Computer-Vision Based Robot Arm | Senior Design May 2025
- Integrated Machine Learning (YOLOv5), OpenCV, and Tensorflow into a 6-axis manipulator for bolt detection and unscrew/screw operations in a radioactive environment.
- Designed and manufactured an end effector that can output 60Nm of torque while maintaining stability.
- Reduced the operation time of unscrewing/screwing bolts onto the flange by 2770%.

Assistive Impedance-Controlled Exoskeleton | Undergraduate Research Apr 2014
- Developed a design that uses force-sensing resistors to detect the force and direction of a user.
- Redesigned and manufactured backplates, braces and linkages that resulted in a 28% weight reduction and increased modularity.

ADDITIONAL

Technical Skills: Advanced in SolidWorks, ANSYS, Python, MATLAB, C++, DFM, ROS2, Gazebo
Certifications & Training: SolidWorks Certificate, Machine Shop trained.
Recommendations: Available upon request.